No. of Functional Features Included in the Solution

Project Title: Streamlining IT Procurement: Automating standard laptop orders with WI

Date	15 March 2024
Team ID	XXXXXX
Project Name	Streamlining IT Procurement: Automating standard laptop orders with WI
Maximum Marks	5 Marks (Feature Documentation Component)

1. Abstract

This project focuses on streamlining the IT procurement process for standard laptop orders inside an organisation by introducing an automated workflow system (called WI – Workflow Integration). Currently, employees raise laptop requests through emails or paper forms. This causes delay, lack of visibility and sometimes budget overruns. The proposed solution digitises the complete journey – from request creation, manager approval, budget check, inventory verification, vendor selection, to automatic purchase order generation and status tracking. The system reduces manual effort, improves compliance and gives real-time status to all stakeholders. All eight planned functional features have been designed, implemented and tested.

2. Introduction

Every company needs to provide laptops to new joiners and also replace old machines for existing employees. In most organisations this process is still semi-manual. An employee sends a request, manager forwards it, finance checks budget, IT team looks for stock or contacts vendor, and finally a purchase order is created. Each step depends on people remembering to act. This leads to long waiting time, lost emails and no clear audit trail.

The goal of this project is to build a simple but complete automated system that handles standard laptop orders using a workflow engine (WI). The system is designed for final-year academic demonstration but follows real-world procurement principles such as request validation, multi-level approval, budget control, inventory first approach and vendor management.

3. Problem Statement

The existing manual / email-based laptop procurement process suffers from the following problems:

- No single place to raise and track a request. Employees keep asking “Where is my laptop?”
- Manager approval is done by email reply, so there is no proper record or SLA tracking.
- Budget is checked only at the end, which sometimes results in last-minute rejection.
- IT team does not always check internal inventory first; they may order new machines even when stock is available.
- Vendor selection is based on personal preference rather than transparent ranking.
- No automated generation of purchase order or delivery notification.

Because of these issues, average turnaround time for a standard laptop can become 15–25 working days. The organisation also loses money through duplicate orders and lack of bulk negotiation.

4. Objectives

The main objectives of the project are:

- To design a simple web-based form for raising standard laptop procurement requests.
- To implement a multi-level approval workflow using WI (manager → finance/budget → IT).
- To automatically check remaining departmental budget before proceeding further.
- To verify available stock in internal IT inventory and prefer internal fulfilment.

- To rank and suggest preferred vendors based on price, delivery SLA and past performance.
- To generate a draft purchase order automatically once all checks are passed.
- To provide a real-time status dashboard and email notifications for every stage.
- To reduce the end-to-end cycle time of a standard laptop order to under 7 working days.

5. Existing System

In the current setup, the process is mostly email and Excel driven:

- Employee writes an email to manager with laptop configuration needed.
- Manager forwards the same email to IT procurement mailbox after adding a note.
- IT team maintains a shared Excel sheet of open requests.
- Budget confirmation is taken by calling or mailing the finance team.
- Vendor quotation is collected manually by phone or separate emails.
- Purchase order is typed in a separate ERP or Word template.
- Employee has no visibility until the laptop is delivered or someone replies.

This system has no automatic routing, no SLA, no audit log and high dependency on individual people. It is difficult to generate monthly reports or find bottlenecks.

6. Proposed System

The proposed system introduces an end-to-end digital workflow for standard laptop orders. Key design decisions are:

- Single entry point – a simple web form for all standard laptop requests.
- Workflow engine (WI) controls the sequence of approvals and automatic actions.
- Role-based access – Employee, Reporting Manager, Finance/Budget Owner, IT Admin, Vendor Viewer.
- Inventory-first logic – system always checks internal stock before creating external order.
- Rules-based vendor ranking – transparent and configurable scoring model.
- Full audit trail – every action (who, when, decision, comment) is stored.
- Email + in-app notifications at every important milestone.
- Configurable SLA and automatic escalation if a step is delayed.

The system is built as a lightweight web application with a workflow backend. It can later be connected to the organisation’s ERP or asset management tool through APIs.

7. System Architecture & Workflow

The solution follows a simple layered architecture:

- Presentation Layer – Web UI for request form, approval screens and dashboard.
- Workflow Layer (WI) – Orchestrates the sequence of steps, timers and escalations.
- Business Logic Layer – Budget check, inventory lookup, vendor ranking, PO generation rules.
- Data Layer – Database storing users, requests, inventory, vendors, budget data and audit logs.
- Integration Layer – Email service and optional future connectors to ERP / asset tools.

Step-by-step Laptop Order Automation Workflow:

Step 1 – Request Submission

Employee logs in, opens “New Laptop Request”, selects standard configuration (or chooses from catalogue), enters justification and delivery location, then submits. System creates a unique request ID.

Step 2 – Manager Approval

WI automatically routes the request to the employee’s reporting manager. Manager can Approve, Reject or Ask for more information. If no action is taken within the SLA (e.g. 2 working days), system escalates to the next level manager.

Step 3 – Budget Check

After manager approval, the system queries the departmental budget table. If sufficient balance exists, the request moves forward and a temporary hold is placed on the amount. If budget is insufficient, the request is sent back with a clear message.

Step 4 – Inventory Lookup

IT inventory module is checked for matching available laptops (same or better configuration). If stock is found, the request is marked for internal allocation and a delivery task is created for the IT asset team. External purchase is skipped.

Step 5 – Vendor Selection

If no stock is available, the system applies ranking rules on the approved vendor list (price score + delivery time score + past performance score). Top recommended vendor is selected automatically; IT admin can override with a reason.

Step 6 – Purchase Order Generation

System creates a draft Purchase Order with all required fields (item, quantity, price, delivery address, cost centre). The draft is sent to the selected vendor by email and also stored in the system.

Step 7 – Order Tracking & Delivery

Vendor confirmation and expected delivery date are recorded. When the laptop arrives, IT team updates status to “Received” and assigns the asset to the employee. The requester receives a final notification.

Step 8 – Closure & Reporting

Request is closed. All data remains available for monthly reports (number of requests, average cycle time, budget utilisation, vendor performance).

8. Functional Features Included in the Solution

This section lists all the functional features that form the complete solution. Each feature is described with its purpose, the module it belongs to, current status and its contribution toward the evaluation marks.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Procurement Request Form	Employee can raise a standard laptop request with required configuration, justification and preferred delivery location.	UI / Frontend	Done	0.6
2	Manager Approval Workflow	Request is automatically routed to reporting manager for approve / reject / request-more-info decision.	Workflow Engine (WI)	Done	0.7
3	Budget Availability Check	System checks remaining departmental budget before sending request to procurement team.	Backend Logic	Done	0.6
4	Vendor Selection & Ranking	Based on price, delivery time and past performance, system suggests preferred vendor from approved list.	Backend / Rules Engine	Done	0.6
5	Inventory Stock Lookup	Checks internal IT inventory first. If matching laptop is available, request is fulfilled from stock.	Database / Inventory	Done	0.5
6	Automated Purchase Order	Once approved and budget confirmed, system generates draft PO and sends it to selected vendor.	Integration / Email	Done	0.5
7	Status Tracking Dashboard	Requester, manager and IT admin can see live status (Pending, Approved, Ordered, Delivered).	UI / Dashboard	Done	0.5
8	Notification & Escalation	Email / in-app alerts at each stage. Auto-escalation if manager does not act within SLA.	Workflow + Notification	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Request, Approval, Budget, Vendor, Inventory, PO)
Additional / Nice-to-Have Features	2 (Dashboard, Escalation)
Features Tested & Verified	8 (Unit + Integration testing completed)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Request Form, Status Dashboard
2	Backend / Logic	2	Budget Check, Vendor Ranking
3	Database / Storage	1	Inventory Stock Lookup
4	API / Integration	1	Automated Purchase Order generation

5	Security / Authentication	1	Role-based access (Employee, Manager, Admin)
6	Workflow Engine (WI)	1	Manager Approval + Escalation

9. Benefits

The automated system provides the following benefits:

- Faster turnaround – expected reduction of cycle time from 15–25 days to under 7 working days.
- Full transparency – every stakeholder can see the current status of any request.
- Better budget control – funds are checked and reserved early in the process.
- Inventory optimisation – internal stock is preferred, reducing unnecessary new purchases.
- Fair vendor selection – ranking is data-driven and auditable.
- Strong audit trail – useful for internal audits and compliance.
- Less manual work for IT and finance teams – repetitive tasks are handled by the workflow.
- Happy employees – they receive laptops faster and can track progress themselves.

10. Conclusion

This project successfully demonstrates how a standard laptop procurement process can be streamlined using a workflow-driven automated system (WI). By covering the complete journey from request to delivery – including manager approval, budget check, inventory verification, vendor selection and purchase order generation – the solution removes most of the manual bottlenecks present in the existing email-based process. All eight planned functional features have been implemented and verified. The design is simple enough for a final-year academic project yet realistic enough to be extended into a real organisational tool.

11. Future Scope

Possible future enhancements include:

- Mobile application so employees can raise and track requests on the phone.
- Integration with the organisation's ERP system for automatic PO posting and invoice matching.
- AI-based recommendation of laptop configuration based on employee role and past requests.
- Support for non-standard / specialised hardware requests with additional approval levels.
- Vendor portal where selected vendors can update delivery status directly.
- Advanced analytics dashboard showing cost trends, vendor performance and forecast demand.
- Multi-language support and multi-currency handling for global organisations.

12. References

1. Laudon, K. C. & Laudon, J. P. – Management Information Systems (latest edition) – chapters on enterprise systems and procurement.
2. ISO 20400:2017 – Sustainable Procurement – Guidance (high-level process principles).
3. Microsoft Power Automate / Workflow documentation – concepts of sequential and parallel approval flows.
4. ITIL 4 – Service Management practices related to request fulfilment and asset management.
5. Academic papers and case studies on digital transformation of procurement processes (open web sources).
6. Internal organisational procurement policy documents (anonymised for academic use).

This document is prepared as original final-year student project work. All feature descriptions, workflow steps and technical points are written specifically for the project titled "Streamlining IT Procurement: Automating standard laptop orders with WI".